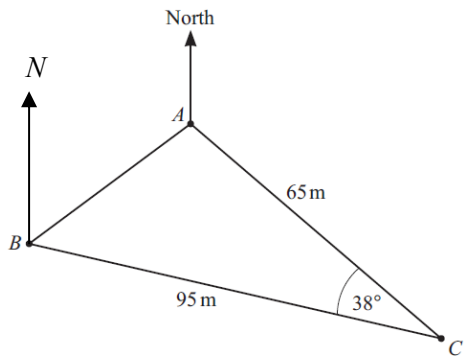
Question	Answer	Marks	Partial Marks	Guidance
17(a)	287	**2**	**M1** for North line at *C* with angle 35 or 145 marked on diagram at *C* isw or for 360 – 35 – 38 oe	M1 implied by answer 73 or 73 marked at *C* as anticlockwise angle from North to *CB* isw
17(b)	$\sqrt{65^2+95^2-2\times65\times95\cos38}$	**M2**	**M1** for $65^2+95^2-2\times65\times95\cos38$ **A1** for 3518[.0] to 3518.1	M2 may be done in steps, allow if square root shown after error provided correct substitution is shown Accept 13 250 – 12 350cos38 as minimum for M2 or M1
	59.31…	**A1**		If M1 and then 59.31…, allow M2A1
17(c)	244.4 to 244.6	**4**	**B3** for [*BAC* =] 99.5 or 99.6 or 99.49 to 99.58… isw or for answer 225.4 to 225.5… OR **M2** for [sin *A* =] $\frac{95\sin38}{59.3}$ oe or [cos *A* =] $\frac{59.3^2+65^2-95^2}{2\times59.3\times65}$ or **M1** for $\frac{59.3}{\sin38}=\frac{95}{\sin A}$ oe or $95^2=59.3^2+65^2-2\times59.3\times65cosA$ oe **M1dep** for 145 + *their BAC* leading to answer	Mark at most accurate Accept answer 244 or 245 For B3 accept answer 225 or 226 if more accurate value seen For M2 or M1 allow more accurate value for 59.3 (condone 59) and allow longer methods M2 implied by [angle *BAC*] = 80.40 to 80.51 M2 for complete method to find angle *A* If longer method and find angle *B* first then angle *B* = 42.4 or 42.42 to 42.44… Allow method for angle *A* if seen in part (a) or by the diagram and used in part (c) Dep on at least M1 See AG

Additional guidance Q17(c)

Alt method – only apply if finding angle *NBA* or angle *NAB*



Question	Answer	Marks	Partial Marks	Guidance
17(c)	244.49 to 244.6	**4**	**M2** for [*NBA* =]180 – 73 – $\sin^{-1}\left(\frac{65\sin 38}{59.3}\right)$oe or for [*NAB* =]73 + $\sin^{-1}\left(\frac{65\sin 38}{59.3}\right)$oe or **M1** for [sin *B* =] $\frac{65\sin 38}{59.3}$ or for [cos *B* =] $\frac{59.3^2+95^2-65^2}{2\times 59.3\times 95}$ **M1dep** for 180 + *their NBA* leading to answer or for 360 – *their NAB* leading to answer	Mark at most accurate Accept answer 244 or 245 For M2 or M1 allow more accurate value for 59.3 (condone 59) angle *B* = 42.4 or 42.42 to 42.44… Dep on M2